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OCR Based Image Text To Speech Conversion Using MATLAB

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Abstract — There are millions of blind people in the world who are visually impaired. Disability to read has a large impact on the life of visually impaired people. The Proposed system is cost-efficient and helps the visually impaired person to hear the text. The main idea of this project is optical Character recognition which is used to convert text character into the audio signal. The text is preprocessed and then used for recognition by segmenting each character. Segmentation is followed by extraction of the letter and resizing of the file containing the text. This Text file is then converted into the audio signal. MATLAB16 will be used for all these processes mentioned above.

Keywords— OCR, Segmentation, Text Extraction, Templates, TTS, MATLAB

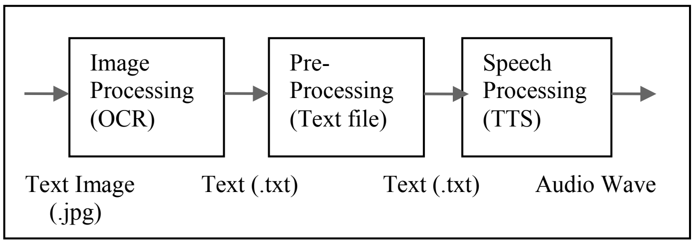

Fig.1. Building blocks of Image to Speech Processing

I. INTRODUCTION

Speech plays a very important role to express one's thoughts. A universal feature of human beings which plays a very important role to establish effective communication. My ultimate aim is to make visually impaired people read books, newspaper; articles etc. braille is the system that already exists to aid the visually impaired people. But the main disadvantage of braille system is that it consumes more time. In braille system, the text is converted to raise dots to make the visually challenged people read and understand the letter /word. The whole designs realize on optical character recognition an image processing Technique to identify each and every character in the text. In this technique, the first image is captured and pre-processing is done by image filtering after that character recognition is done. This is followed by audio conversion. MATLAB16 is used for all these processes. The beloved figure shows the basic blocks of the image to Speech Processing Technique. The first block the text is extracted from the input image using OCR then the extracted text is converted into Speech using TTS in MATLAB.

II. LITERATURE SURVEY

Kumara et al [1] the paper describe the implementation of text is extracted and it converted into speech using Raspberry Pi.

Chaudhari et al [2] the paper describe the implementation of Marathi voice synthesis using Raspberry Pi, Hm 2007, Motor, L239D and Ultrasonic Sensors.

Chandran et al [3] the paper describes the implementation of speech generation system using Matlab, Win 32 SAPI software (library) and creates a database.

Gopinath et al [4] the paper describe the image is converted into text then the text is converted into speech using Matlab, lab view and android platform.

Badhe et al [5] the paper describes Marathi script (Devanagari script) the TTS system has been performed on Matlab R2014 using Unicode software and UTF-8 is used for reading the Marathi language in text format and also creating the audio database, text database.

Patil et al [6] the paper describe the implementation of TTS using OCR using Matlab and Microsoft win 32 SAPI software.

Kaladharan et al [7] the paper describe the Text is converted into speech using C#, and Microsoft.NET from work 4.5 SDK using Microsoft Visual C#.

Bansal et al [8] the paper describe TTS conversion using Matlab and Microsoft SAPI software.

Anuradha et al [9] the paper describe developing a pc based text to speech synthesizer using phonemes, concatenation, and prosody using Matlab.

Savadisvar et al [10] the paper describes text is covered into speech using the I Novel algorithm using Matlab and use beagle board and Arduino voice shield box TTS256 processor.

III. PROPOSED WORK

A. *Optical Character Recognition*

OCR is a reader which recognizes the text character of the computer which may be in printed or written form. the OCR templates of each character are used to recognize the character i.e. scanning process is carried out. After this, the character image is translated in ASCII code which is further used in data processing.

OCR follows some basic architectural framework module as we can see it in figure 2 given below. It thresholds with the image recognition where each character from the image is being sent for recognition. There are some pre-processes involved to make the image noise free which involves process like binarization, skew correction and normalization [8]. Where we can see the image undergoing some enhancements such as filtering out noise and contrast correction. After this, the real framework of OCR starts with the segmentation process, as the name says everything here the characters is segmented so that they can be texted separately. At the next level of the framework the segmentation is followed by the text lines, words and characters in the image. This level is assisted by some connected component analysis Information and projection analysis to again assist text segmentation. Further the process is followed by feature extraction which is concerned with the representation of the object. All the real-time working on the OCR lies in the Feature extraction which can be titled as the 'Heart' of OCR.

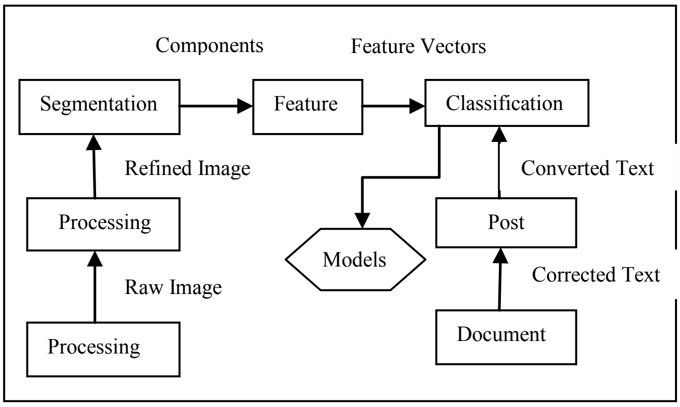

Fig.2. OCR Framework

B. *Text Recognition*

The initial process is to capture the image using a webcam/camera followed by the text preprocessing. When the text area is been taken in the circumstance the internal framework of the processes starts. The text is segmented then those recognized texts are rebuilt in the original text. This information is used to reconstruct the numbers and words in the original text. For recognition of captured characters, the flow of the proposed work goes as follows binarization, optical scanning, feature extraction, segmentation and character recognition.

1. *Image is captur*ed

The Image is captured through webcam and saved.

2. *Binarization*

A process of binarization is converting a grayscale image into a binary image using thresholding is known as Binarization. Before making the phenomenon acknowledged taking you the decades back, it was used in faxes now the binarization is really easy but typical to understand in simple words we know that the image contains pixels which are stored bit by bit, now in image there are two colours black(0)and white(1) what does it do is that pixels which are grey it makes them set(accepted) and the pixels with white are made unset further in the process the pixels which are in set mode and are near are combined to make some acceptable character. The important characteristic of the binarization is the distance transformation by which the unset pixel is distanced from another set pixel.

3. *Segmentation*

Segmentation is done as the image consists of a number of sentences and each line contains a certain number of the words. Then this word is formed by the number of characters. Hence we can say that the segmentation is a process of partitioning the digital image into the segments. In the segmentation process each line, each word, each character is segmented. There are some inevitable difficulties in segmentation process like image quality is less, every computer system has some different fonts, cursive writing, etc. this affects the efficiency of the segmentation process thus the best practice is done here to eliminate those problems.

4. *Feature Extraction*

In the feature extraction, the main work is to extract the essential characteristics of the character generated by joining the set pixels. Probably guessed and accepted that the inevitable problems occur is in the pattern recognition. .mostly the description of the character is carried out by its actual raster image. The extraction feature is carried out in five different ways:

1. Correlation techniques and Template Matching.
2. Techniques of Feature-based.
3. Points Distribution.
4. Series expansions and Transformations.
5. Analysis of Structural.

5. *Recognition*

Ones the character is segmented its stored in a variable as to compare it with the stored template form. Precisely preliminary data will be stored in the form of templates where all characters recognized font and size are available. The data contains further information: the value of ASCII character, the name of the character, character of the image in Jpg and character width, etc. for every identified character all information will be captured then there is the comparison between the recognized character information with the predefined character which is stored in the system. As used same font and size for the identified character extract one unique match. There is variation in the size of the character and then it is scaled with the known standard thus completing the recognition process.

C. *Text to Speech Conversion*

We know that the artificial production of human sound is nothing but the Speech. Some mechanism is used to generate speech this mechanism is known as a speech synthesizer. In that, we can give the sound of human and robot also but generally, we use the robot. The text to speech system is divided into two sections: namely a back-end and front-end. The front-end has two vital tasks. Now the process starts with the conversion of the raw text like abbreviations, symbols, numbers and it a little bit similar to written out words.

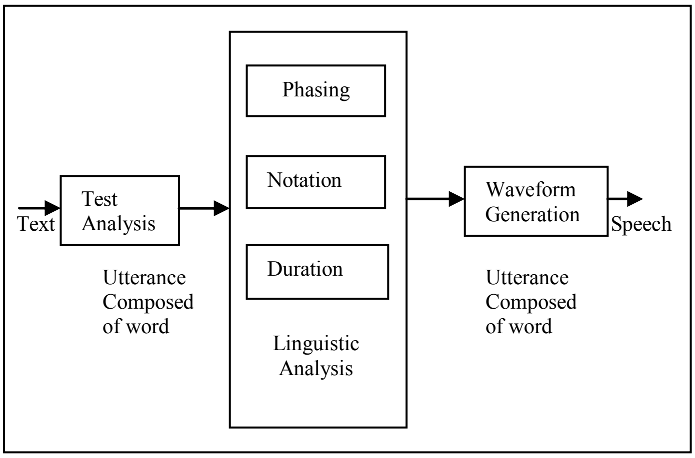

Fig.3. TTS system

This process is as text preprocessing, normalization. Now by assigning the phonetic transcription of the front ends of every word and divides the text into clauses, prosodic units, phrases and sentences. The phonetic transcription is assigned into the word is known as the text is converted into phoneme or graphene is converted into phoneme. The output of the front end is nothing but the prosody information and phonetic transcriptions of the symbolic linguistic representation. Whereas back-end is converted into symbolic linguistic representation into Audio form, hence also referred as the synthesizer.

IV. PROJECT IMPLEMENTATION

The proposed approach steps are as follows:

1. Step: The image is captured using a webcam and it is stored in the form of (.jpg) file format. Next step image is read/display using the imread command. It will read the image from stored file.

2. Step: The preprocessing conversion of the original RGB image into a grayscale is done by rgb2gray command. In this, as discussed above the pixels are made set and unset. The commands Rgb2gray it convert RGB image or colormap to grayscale image. Rgb2gary conversion RGB image to grayscale by removing the hue and the saturation information which is Re-training the luminance. The Command fid = open (filename) open the file using filename contains the name of the file to be opened. And characters are stored in an empty matrix.

3. Step: The image filtering. Filtering the unwanted noise is removed which smoothing the image for further processing.

4-10. Step: The process performs segmentation of each and every line, every letter. A correlation process in which the file is loaded so as to match the letter into stored templates.

11. Step: The next step is the conversion of text to speech. The first analysis, text and it will convert into speech using MATLAB.

12. Step: Finally get the speech for given image.

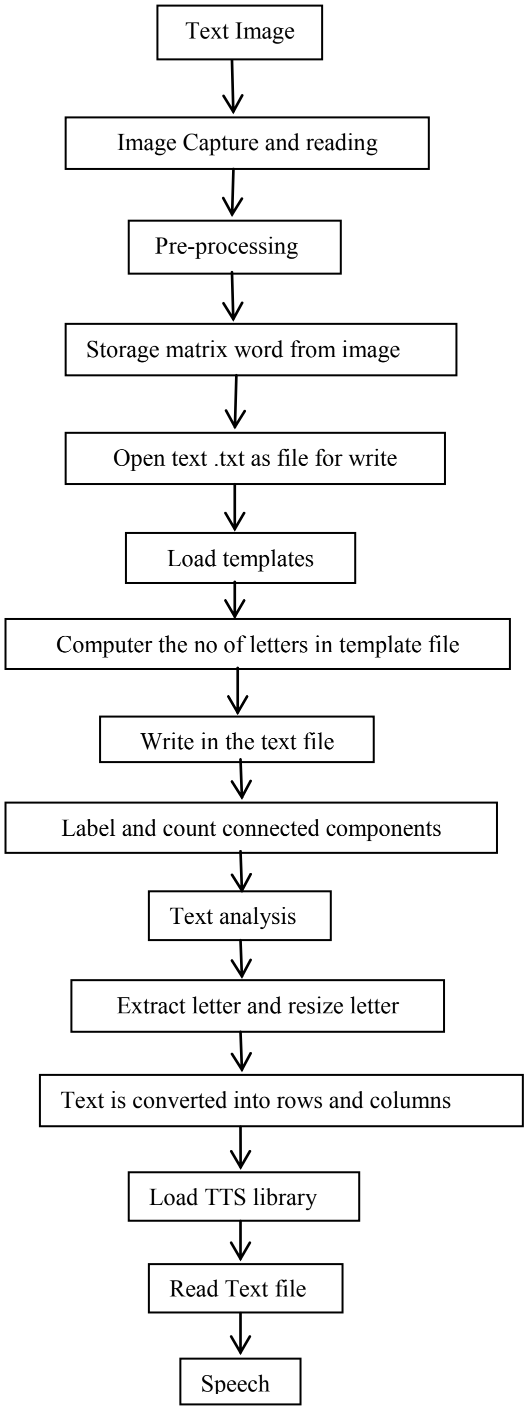

Fig.4.Flowchart

V. RESULT

The Text to speech system consists of two parts: First text is recognizing using OCR and second is text to audio conversion using MATLAB. It is a real-time system first capture the image through the webcam and it converts the text file then the text file is coveted into the speech. That is shown in figure 5 and 6 below.

The advantage of this system does not need the internet connectivity that is a basic necessity for other TTS system like a Google TTS in Google keep.

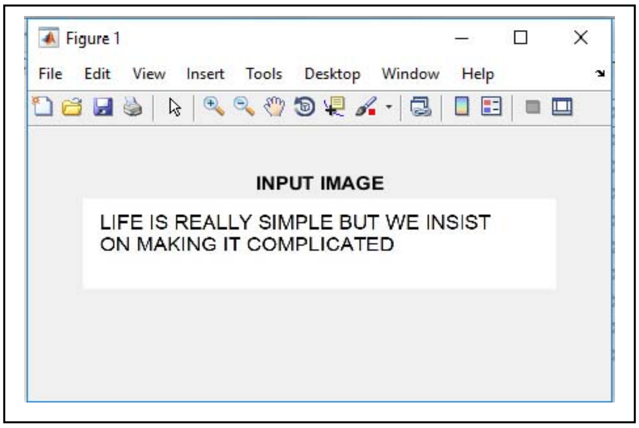

Fig.5.Input Text Image

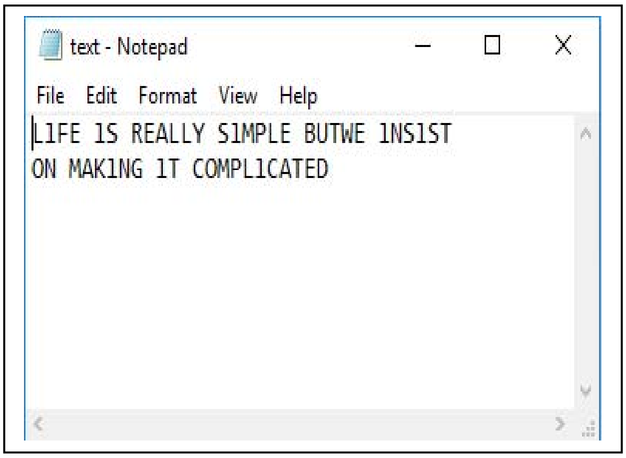

Fig.6.Output Text Image

VI. CONCLUSION

This paper reflects effort being carried out for the text is extracted from the image using optical character recognition and the recognized character is converted to audio using MATLAB environment. This application proposed is cost-effective, user compatible and real-time system. By this system, read a text file from documents, newspaper, E-mail. This system can also be advantageous for people who are visually handicapped. And the more interesting fact is I'm really trying to make one hardware space where MATLAB. But the program can be installed and can be carried everywhere easily. The main purpose of this system is to fulfill the needs of the People who are vocally handicapped and also can have the communication with people who do not understand the sign language. Another help is the one can learn the pronunciation of the words correctly. TTS system also can be used in domain-specific applications such as train announcements. The system can be used to build information surfing for the people who can't write and read. My project work may help blind people to read the documents and books deaf will get benefited to share their feelings /opinion. The practical experiment is performed text reading system and better results are obtained .at last project seems to be easy but the real-time problems where really inevitable like making templates of every character with every font and making it real time.

VII. FUTURE SCOPE

This implementation of the image to speech processing can be implemented on some standalone device. This can be very beneficial for the visually impaired people. And if use the translation block just after the word correction block, it can also be used as translator device that can read image having text and translate it into multiple languages.

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